

CSC902: Advanced Algorithms

Preliminaries: Asymptotic Notation, Algorithm Paradigm: Greedy, Divide and Conquer, Master Theorem, Dynamic Programming (Chained Matrix Multiplication). Review of Mathematical Background: Recurrence Relation, Generating Function.

Algorithmic Computation: Family of Algorithms, Number-theoretic Functions, Primitive recursion.

Advanced data structure: Skip list, Union find, 2-3-4 trees, red-black tree, splay tree, Tries and suffix trees, Universal Hashing.

Word-level Parallelism. Transdichotomous Model.

Graph Algorithms: All pair shortest Path, Network flow, Ford Fulkerson method, Matching.

String matching algorithms: KMP and Rabin-Karp.

Number theoretic algorithms: Modular Arithmetic, Primality Testing, Chinese Remainder theorem. Polynomial representation, DFT and FFT.

Geometric algorithms: convex hulls, line segment intersection, closest pair.

Randomized Algorithms: concept, problems.

Non-determinism: NP-completeness, Co-NP, P Space.

Fixed parameter algorithms: Parameterized Complexity, Kernelization for vertex cover.

Approximation Algorithm: Performance bounds, vertex cover problem, Travelling salesman problem, Greedy Set Cover problem, Linear Programming Relaxations, Randomized Rounding.

On-line algorithms: competitive ratio, list accessing, paging, self-organizing data structures, k-server, networks, auction, load-balancing, lower-bounds, Ski Rental, River Search Problem, List Ordering and Move-to-Front.

Streaming algorithms: Streaming Data models: Distinct items, frequent items, frequency moments, estimating norms, clustering.

Parallel Algorithms: PRAM, Selection, Search, Merge, Sort algorithms. Pointer Jumping and Parallel Prefix, Tree Contraction, Divide and Conquer, Randomized Symmetry Breaking, Maximal Independent Set.

Reference Books:

1. Jon Kleinberg and Éva Tardos, *Algorithm Design*, Addison-Wesley, 2005.
2. Thomas Cormen, Charles Leiserson, Ronald Rivest, and Clifford Stein, *Introduction to Algorithms*, Third Edition, MIT Press, 2009.
3. Sanjoy Dasgupta, Christos Papadimitriou, and Umesh Vazirani, *Algorithms*, McGraw Hill, 2006.
4. *Introduction to Computability*, F C Hennie, Addison Wesley.
5. *Discrete Mathematics*, Johnsonbaugh Maxwell Macmillan.